



Design, identification, and control of a linear dual-stage actuation positioning system[☆]

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Abstract

Voice coil actuators (VCAs) are widely used for motion control applications that require high precision and fast acceleration. However, they have one disadvantage of short travel ranges typically on the order of centimeters. To overcome this disadvantage, this paper employs an existing concept called dual-stage actuation and combines a linear VCA with a low-cost dc-motor (DCM) to construct a linear dual-stage actuation (DSA) positioning system, which extends the travel range as large as that provided by the DCM while retaining the positioning accuracy offered by the VCA. For this setup, proper controllers are crucial to achieve the desired performance. Hence, we first identify the DSA model and investigate the interaction force between the VCA and DCM stage. Next, robust controllers are specifically designed for the DCM and VCA stage, respectively. Finally, experiments are carried out to verify the robust tracking performance of the developed linear DSA positioning system. It is demonstrated that the developed DSA positioning system offered a cost-effective solution to extending the working range of the existing VCA whilst retaining the positioning accuracy.

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1. Introduction

Voice coil actuators (VCAs) are characterized by compact size, high force-to-weight ratio, and no cogging ripple or commutation. These advantages enable the VCAs ideal for control applications where positioning performances of high speed and high precision are essential. According to the type of motion, the VCAs can be manufactured as rotary or linear ones based on the same operation principle. Both types of VCA have been widely used in industrial motion control systems. For example, the rotary VCA is commonly used in hard disk drives (HDDs) nowadays to position the read/write head over the disk surface [1]. Meanwhile, the linear VCAs are adopted to develop a low-cost ultrasound scanner [2] and to move the radial position of the objective lens in optical disk drives (ODDs) [3].

However, the main disadvantage of most commercial VCAs is their short travel ranges, which are typically on the order of a few centimeters for linear ones and a few ten degrees for rotary ones. As a result, the VCA is hardly suited for applications that need both a large operation range and high accuracy such as surface inspection and pick-and-place assembly. Motivated by this, we develop a new linear positioning system by using an existing concept of dual-stage actuation (DSA) proposed in [4] to considerably extend the operation range of the VCA. Specifically, we design a low-cost dc-motor (DCM) actuated stage which is capable of carrying the VCA stage to move throughout a much larger operation range in the same axis as the VCA, irrespective of its coarse positioning accuracy. By simultaneous control of the DCM and VCA stage in response to reference commands, the combined system can provide both an extended operation range dominated by the DCM stage and high positioning accuracy determined by the VCA.

Although the mechanical configuration of DSA systems is straightforward, the combination of the two actuators that can drive each stage properly differs depending on the applications. For instance, in dual-stage HDDs the VCA is used as the coarse actuator while a much smaller piezoelectric actuator (PEA) is commonly used for fine positioning [4,5]. The DSA machine tool that needs high torque output can be formed by an electrohydraulic actuator and a large PEA [6]. Whereas, the ODD mechanism is typically comprised of a coarse dc-micro-motor (or a stepping motor) and a fine VCA to move the laser spot on track despite that the control of the two actuators is sequential rather than simultaneous which may not enhance the response speed [7]. For DSA systems with either stage driven by a different actuator, the control design should be carefully investigated not only because the control problem for either stage is different but also because the new actuator dynamics may result in different interactions between the two stages. As such, there have a variety of approaches reported for the DSA control such as the literature in [8–10] and references therein.

In this paper, we will first present the system identification of the developed DSA system as well as the analysis of its interaction effect. Next, the control design for each actuator is addressed in details to solve the control problems as formulated. In particular, a robust controller based on fast non-singular terminal sliding mode (FNTSM) [11] is employed for the VCA stage to cope with the disturbances and system uncertainties such as payload

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variations. The FNTSM controller can guarantee exponential stability and fast finite-time convergence of the closed-loop system [12]. Moreover, it involves no switching elements and is thus free of control chattering for practical implementation. These features enable the FNTSM control to outperform the linear proportional-integral-derivative (PID) control [13] and conventional sliding mode control methods [14,15], which have been experimentally demonstrated on motion control systems driven by a linear motor [16] or a piezoelectric actuator [17], and underwater vehicle [18]. We also discuss the selection criteria of the controller parameters taking account of the actuator dynamics and the design compromises including measurement noise and relative position of the two stages for avoiding VCA output saturation. Finally, we present the experimental results to verify the superior performance of the controllers in terms of more robust and accurate tracking and faster disturbance compensation over the extended operation range compared with a linear H_∞ controller.

2. Design of a linear DSA positioning system

Our design objective is to extend the applicable travel range of a linear VCA stage while maintaining its advantage of high positioning accuracy. Intuitively, this goal can be achieved by using the structural configuration where the VCA stage is carried on the back of a DCM driven stage which has a larger travel range along the common axis. By coordinating the two stages with properly designed servo controllers, the configuration actually becomes a linear DSA positioning system which simultaneously provides the merits of each stage, that is, a large travel range, high positioning accuracy and fast response.

Fig. 1 shows a prototype of the linear DSA positioning system, which consists of a DCM stage with a piggyback VCA stage. Both stages are supported by linear ball bearing rails with low friction and moving along a common axis. The travel ranges of the DCM and VCA stage are 400 mm and 20 mm, respectively. Accordingly, the workpiece can be placed over the VCA stage and its linear position (i.e., our control object) is aimed at reaching an extended range as provided by the DCM stage and achieving the positioning accuracy as offered by the VCA stage.

From Fig. 1(b), we can see that the DCM stage is driven by a low-cost DC motor through a pinion-and-rack mechanism that converts rotary motion to linear one. The DC motor is integrated with a gear box to increase the driving force. Ideally, the DCM stage can provide unlimited travel range provided that its linear bearing rails are in sufficient length. However, the disadvantages are that both extra frictions from the gear box and backlash effects in the pinion-and-rack mechanism would deteriorate the positioning accuracy. Nevertheless, these disadvantages will be effectively compensated by the VCA in our DSA setup.

On the other hand, the VCA stage is directly driven by a VCA (by H2W Technologies) which is composed of two parts: one moving coil holder which is firmly attached onto the sliding stage; and one permanent magnetic cylinder which is mounted on the DCM stage. Apparently, the travel range of VCA stage is largely limited by the length of the coil winding. Moreover, the non-contact between the coil and the magnetic cylinder avoids additional friction. However, one arising control problem is that the relative position between the VCA stage and the DCM stage (i.e., the relative position of the moving coil and the magnetic cylinder) should be guaranteed within the coil length to avoid VCA output saturation. The DCM and the VCA are driven by a linear power amplifier, respectively. Each amplifier can be simply modeled as a constant gain due to its relatively higher bandwidth than the correspond-

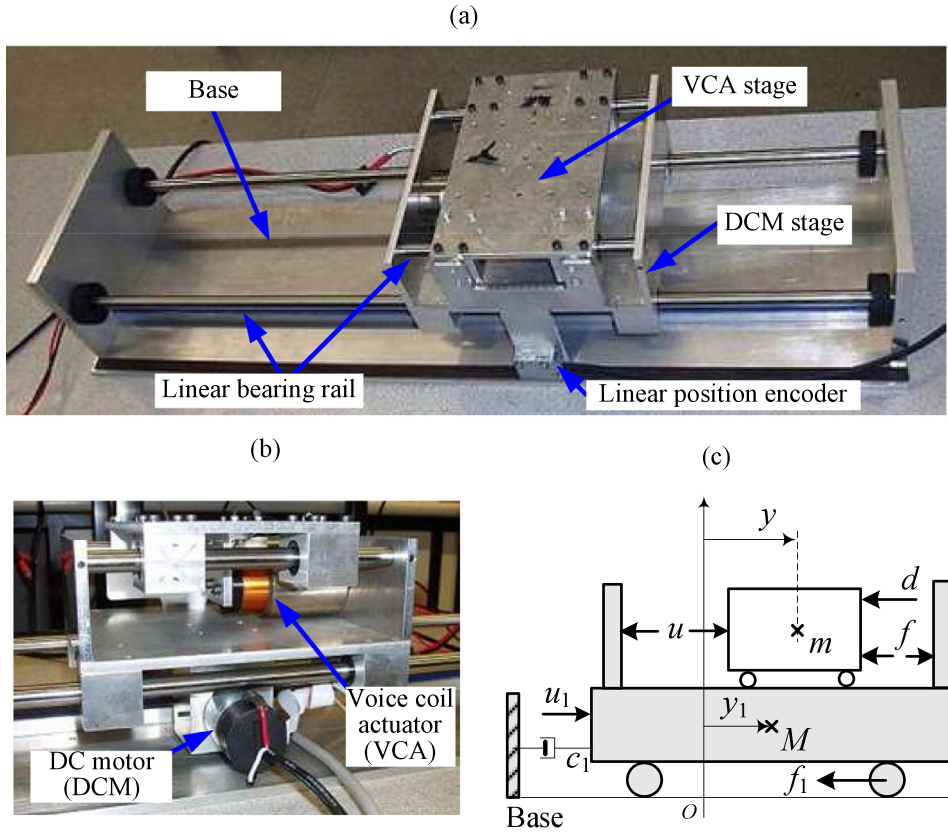


Fig. 1. Prototype of a linear DSA positioning system. (a) Top view; (b) Rear view; (c) Illustration of the DSA configuration.

ing motor dynamics. For the ease of representation of the control design, we assume each amplifier gain as unity from now on, which is, however, scaled up in the real implementation.

For the purpose of feedback control, it can be seen from Fig. 1(a) that the position of the VCA stage relative to the base (i.e., the absolute DSA position output) is measured by a linear position encoder with a resolution of $1 \mu\text{m}$ (by Renishaw PLC). Meanwhile, the DCM assembly has a shaft encoder with a translated resolution of $5 \mu\text{m}$ along the linear axis. Note that we intentionally choose coarse sensor and motor dynamics in the DCM stage compared to the VCA stage not only for reducing the cost but also for the fact that the DCM stage is employed merely for providing a large travel range rather than high positioning accuracy.

Lastly, Fig. 1(c) illustrates a physical model of the overall DSA configuration, where m and u denote the mass (including payload) and control input of the VCA stage, respectively. y denotes the absolute position of the VCA stage from the origin O along the horizontal axis. M , u_1 , and y_1 denote the counterpart of the DCM stage; and c_1 is the damping coefficient associated with the DCM. f_1 indicates the frictions rising from the DCM assembly as well as the contact between the DCM stage and its linear bearing rails. Similarly, f indicates the friction between the VCA stage and its linear bearing rails mounted on the DCM stage. d is the external disturbance. Note that the control input u and the friction force f when applying

to the VCA stage, will act as counterforce resulting interaction with the motion of the DCM stage. This effect as well as the DSA plant modeling will be analyzed in the next section.

3. System identification

According to Fig. 1(c), the dynamic equation of the DSA system is given by

$$M\ddot{y}_1 + c_1\dot{y}_1 = u_1 - u + f - f_1 \quad (1)$$

$$m\ddot{y} = u - f - d \quad (2)$$

where m denotes the total mass of the VCA stage plus the payload if any; and the friction forces f and f_1 are simply modeled as the following equations:

$$f_1 = \bar{f}_1 \text{sgn}(\dot{y}_1) + \Delta_1 \quad (3)$$

$$f = \bar{f} \text{sgn}(\dot{y} - \dot{y}_1) + \Delta \quad (4)$$

where $\bar{f}_1$ and $\bar{f}$ represent the Coulomb friction level and Δ_1 and Δ are the unmodeled friction, respectively. In particular, the friction f is a function of the relative velocity of the two stages. Note that in this paper we are not aiming for an accurate friction model because it is time varying with respect to various payloads applied. Furthermore, the discrepancy of the friction model can be likely regarded as uncertainty which can be effectively compensated by the robust controller as will be discussed later.

The model parameters in Eqs. (1)–(4) can be either extracted from the mechanical design and fabricating process or identified through experimental data. First of all, we set the VCA stage without payload. Then, the masses of the stages are easily weighed as

$$M = 1.8 \text{ kg}, \quad m_0 = 0.6 \text{ kg}$$

where m_0 denotes the mass of the VCA stage without payload. Next, the remaining model parameters are identified through experiments. We rigidly fix the VCA stage on the DCM stage to avoid any relative sliding motion that means $y \equiv y_1$. Subsequently, adding Eq. (1) to Eq. (2) where m reduces to m_0 and neglecting the disturbance forces f_1 and d leads to the transfer function of the DCM stage as

$$P_1(S) = \frac{y_1}{u_1} = \frac{1}{(M + m_0)S^2 + c_1S} \quad (5)$$

where S represents the Laplace operator. Similarly, from Eq. (2) we can straightly obtain the transfer function of the VCA stage as

$$P(S) = \frac{y}{u} = \frac{1}{m_0S^2} \quad (6)$$

In order to identify the parameter c_1 and verify the accuracy of the models, a dynamic signal analyzer (HP 35670A) is used to generate the swept-sinusoidal excitation signals and collect the experimental frequency response data from the excitation signals to the output. After that, the value of uniquely unknown parameter c_1 can be easily found out equal to

$$c_1 = 120.64 \text{ Ns/m}$$

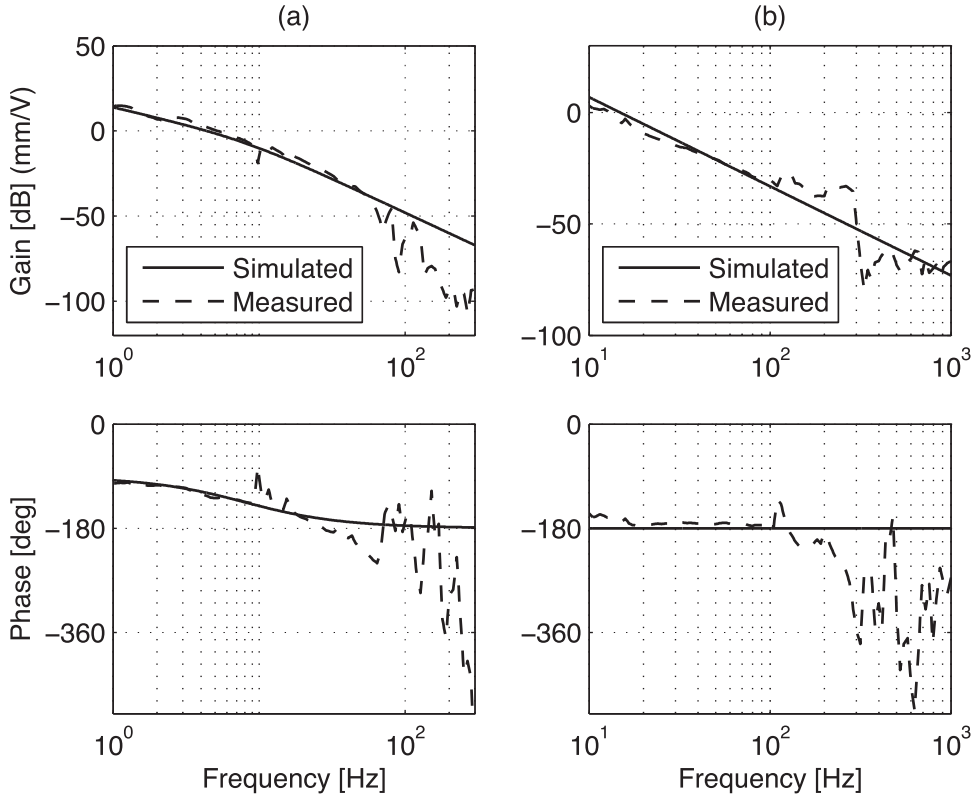


Fig. 2. Frequency responses of the DSA model. (a) DCM stage; (b) VCA stage.

by matching the simulated model with the measured one. The dashed lines in Fig. 2 show the measured frequency responses of the DCM and VCA stage. It can be seen that the simulated results and measured data for DCM dynamics matched well in a clear region within 40 Hz, above which the complex and uncertain motor dynamics and sensor noises deteriorate such a matching phenomenon. Comparatively, the VCA dynamics has a wider matching region of simulated results and measured data within 100 Hz, which is dominated by the rigid body mode [i.e., the double integrator model in Eq. (6)]. The solid lines in Fig. 2 show that the simulated models with the identified parameters match the measured models well in the frequency range of interest.

Finally, by following the experimental procedure in [19] the friction parameters $\bar{f}_1$ and $\bar{f}$ in Eqs. (3) and (4) can be respectively estimated as

$$\bar{f}_1 = 7.06 \text{ N}, \quad \bar{f} = 1.76 \text{ N}$$

when no payload is applied to the DSA stage.

It can be seen from Eq. (2) that the VCA stage is essentially decoupled from the control force u_1 of the DCM stage. Nevertheless, the DCM stage is disturbed by the control force u of the VCA stage as shown in Eq. (1). In order to investigate the extent of this interaction, we carry out an experiment where the VCA stage is controlled to travel across its maximum working range (i.e., 20 mm) under a linear H_∞ feedback controller (This controller will be

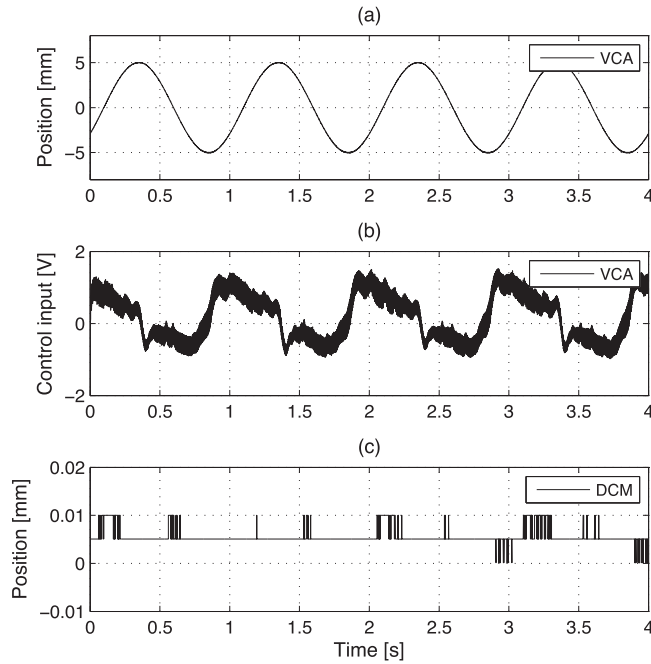


Fig. 3. Interaction effects of the VCA motion on the DCM stage. (a) Position profiles of the VCA stage under a linear controller; (b) Control inputs of the VCA; (c) Perturbed position of the DCM stage.

given in Section 5.3 later). Fig. 3(a) and (b) depict the position profiles and the corresponding control forces of the VCA stage, respectively. In the meanwhile, the DCM stage is set without control (i.e., $u_1 = 0$) and thus its motion is only governed by the control force u according to Eq. (1). From the results in Fig. 3(c), it is interesting to see that the position of the DCM stage is slightly disturbed by 0.01 mm, which is trivial relative to its desired positioning accuracy. The reason for this negligible interaction force is mainly due to $M \gg m_0$. Therefore, we can reasonably ignore the dynamic interaction from the VCA to DCM stage and simply consider its effect as a bounded disturbance for the ease of control design.

4. Control design

This section first formulates the control problem of the identified DSA system. Next, robust control design methods are presented for the two stages, respectively. Finally, we analyze the condition for the two stages to cooperate with each other in order to avoid the VCA from output saturation.

4.1. Problem formulation

Based on the aforementioned analysis, we shall rewrite the DSA model in a compact form as follows:

$$M\ddot{y}_1 + c_1\dot{y}_1 = u_1 - \delta_1 \quad (7)$$

$$m\ddot{y} = u - \delta \quad (8)$$

where $\delta_1 = f_1 + u - f$, $\delta = f + d$, and m are, respectively considered as disturbances and uncertainty in our application and their values are bounded by

$$|\delta_1| \leq \bar{\delta}_1 = 20 \text{ N} \quad (9)$$

$$|\delta| \leq \bar{\delta} = 20 \text{ N} \quad (10)$$

$$\frac{1}{\tau} \leq \frac{m}{m_0} \leq \tau = 2.7 \quad (11)$$

Here, τ represents the ratio of uncertain mass (e.g., a maximum payload of 1 kg) relative to nominal mass of the VCA stage. From now on, we will regard the equations in Eqs. (7)–(11) as the plant models for control design purpose.

The control objective for the DSA system is then to design the control inputs u_1 and u such that the VCA output y can follow a reference command y_r of amplitude beyond the travel range of the VCA stage. Moreover, the tracking performance should be robust against the disturbances δ_1 , δ and uncertainty associated with m . Apparently, the control objective imposes the design requirements that the DCM output y_1 should follow y_r as accurate as possible and also the relative position of the DCM and VCA stage as defined by Δ_y satisfies the following condition to avoid the VCA from output saturation:

$$|\Delta_y| = |y - y_1| \leq L_v \quad (12)$$

where $L_v = 10 \text{ mm}$ denotes half of the maximum travel range of VCA relative to the DCM stage.

4.2. Control design for the DCM stage

As aforementioned, the DCM stage is mainly used for extending the travel range and its coarse positioning accuracy is tolerable from the point of view of the VCA stage. Hence, we directly employ a traditional linear H_∞ design method [20] for the control of the DCM stage. Here, we only give a brief description of the design procedure.

Define the tracking error of the DCM stage as

$$e_1 = y_1 - y_r \quad (13)$$

where y_r is the reference command supposed to be twice differentiable. Let the control input u_1 have the form of

$$u_1 = c_1 \dot{y}_1 + M(\ddot{y}_r - \alpha_1 \dot{e}_1 - \alpha_2 e_1 + u_d) \quad (14)$$

where α_1 , α_2 are the control parameters to be designed and u_d the auxiliary control input further designed to deal with the disturbance. Substituting Eq. (14) into Eq. (7) yields

$$\ddot{e}_1 + \alpha_1 \dot{e}_1 + \alpha_2 e_1 = u_d - \frac{\delta_1}{M}. \quad (15)$$

Hence, we can first choose appropriate values of α_1 and α_2 using the pole placement design method [13] such that the following error dynamic equation of the DCM stage

$$\ddot{e}_1 + \alpha_1 \dot{e}_1 + \alpha_2 e_1 = 0 \quad (16)$$

in the absence of disturbance achieves the desired response. Then, by using the H_∞ technique the auxiliary control input u_d is chosen as

$$u_d = Hx \quad (17)$$

where $x = [e_1 \ \dot{e}_1]^T$ and the gain vector H is defined by

$$H = -\frac{1}{\rho^2} B^T P \quad (18)$$

with $P = P^T > 0$ being a solution to the following Riccati inequality

$$A^T P + PA - \frac{1}{\rho^2} P B B^T P + Q < 0 \quad (19)$$

where

$$A = \begin{bmatrix} 0 & 1 \\ -\alpha_2 & -\alpha_1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and $\rho \in (0, 1)$ and $Q = Q^T > 0$ are respectively, a prescribed scalar and weighting matrix. Following the proof in [20], we can show that there exists a finite positive ϵ such that

$$|e_1| \leq \rho^2 \left(\frac{\bar{\delta}_1}{M} \right)^2 + \epsilon. \quad (20)$$

This implies that the H_∞ controller in Eq. (14) can guarantee an optimal bounded tracking error in the presence of the disturbance. Furthermore, a smaller value of ρ will lead to smaller tracking error but inversely result in larger magnitude of the control input.

By taking into account the linear dynamic bandwidth of the DCM as shown in Fig. 2(a), we choose $\alpha_1 = 119$ and $\alpha_2 = 3948$ which leads to a closed-loop servo bandwidth of approximately 10 Hz for the DCM stage as indicated by Eq. (16). As a result, the following H_∞ controller

$$u_1 = 1.8\ddot{y}_r + 120.6\dot{y}_1 - 7.1 \times 10^3 e_1 - 215\dot{e}_1. \quad (21)$$

is obtained for the DCM stage.

4.3. Control design for the VCA stage using robust FNTSM

Compared to the DCM stage, the control design for the VCA stage is required to maintain the positioning accuracy robustly against the likely larger disturbances and payload uncertainty. The traditional linear H_∞ control may not afford the desired performance under this circumstance. Therefore, we employ a nonlinear robust controller using the FNTSM [16] for the VCA stage.

Let the tracking error of the VCA stage be

$$e = y - y_r. \quad (22)$$

In addition, define a sliding variable s as

$$s = e + \lambda \text{sig}(\dot{e})^\gamma \quad (23)$$

where $\lambda > 0$, $1 < \gamma < 2$, and the notation $\text{sig}(z)^a$ is defined by

$$\text{sig}(z)^a = |z|^a \text{sgn}(z). \quad (24)$$

Note that the function $\text{sig}(z)^a$ for $a > 0$, $\forall z \in \mathbb{R}$ is smooth and monotonically increasing and always returns a real number. The benefit resulting from Eq. (23) is that the corresponding terminal sliding mode function

$$s = e + \lambda \text{sig}(\dot{e})^\gamma = 0 \quad (25)$$

for any initial conditions of $e(0)$ and $\dot{e}(0)$ can converge to zeroes in a finite time t_s [21] given by

$$t_s = \frac{\lambda^{-\frac{1}{\gamma}}}{1 - \frac{1}{\gamma}} |e(0)|^{1 - \frac{1}{\gamma}}. \quad (26)$$

Next, we will derive an equivalent control input [14] expressed by u_e that would maintain

$$\dot{s} = \dot{e} + \lambda \gamma |\dot{e}|^{\gamma-1} (\ddot{y} - \ddot{y}_r) = 0 \quad (27)$$

assuming the plant model is free from disturbances and uncertainties. More specific, let $m = m_0$ and $\delta = 0$ in Eq. (8), and replace u by u_e . Then, Solving Eq. (27) for u_e by using the nominal plant model yields

$$u_e = m_0 \ddot{y}_r - \frac{m_0}{\lambda \gamma} \text{sig}(\dot{e})^{2-\gamma}. \quad (28)$$

Lastly, in order to provide robustness we introduce a reaching control input u_r as follows

$$u_r = -m_0 [g_1 s + g_2 \text{sig}(s)^\beta] \quad (29)$$

where $g_1, g_2 > 0$, $0 < \beta < 1$, and the sliding variable s is with Eq. (23).

In summary, by employing the results in [16] we can state that for the VCA system in Eq. (8) with the uncertainties in Eqs. (10) and (11), if the VCA system is under the FNTSM controller given by

$$u = u_e + u_r \quad (30)$$

where u_e, u_r is with Eqs. (28) and (29), respectively, then the following tracking performance for the VCA closed-loop system can be guaranteed

(i) The sliding variable s converges to the region of

$$|s| \leq \Phi = \min(\Phi_1, \Phi_2) \quad (31)$$

$$\Phi_1 = \frac{(\tau - 1) |\ddot{y}_r - \frac{1}{\lambda \gamma} \text{sig}(\dot{e})^{2-\gamma}| + \frac{1}{m_0} \bar{\delta}}{g_1} \quad (32)$$

$$\Phi_2 = \left[\frac{(\tau - 1)|\ddot{y}_r - \frac{1}{\lambda\gamma} \text{sig}(\dot{e})^{2-\gamma}| + \frac{1}{m_0} \bar{\delta}}{g_2} \right]^{\frac{1}{\beta}} \quad (33)$$

in a finite time, where $\bar{\delta}$ is given in Eq. (10).

(ii) The tracking error e converges to the region of

$$|e| \leq 2\Phi \quad (34)$$

in a finite time.

Remark 1. The FNTSM control law is inherently continuous which means that it is immune from chattering and singularity for practical implementation on a physical system. Furthermore, the FNTSM has the property of exponential stability which implies fast convergence rate in response to external disturbances. These benefits are significant in comparison with other kinds of sliding mode controllers such as [11,21].

In the following, we shall discuss the selection guideline of control parameters of FNTSM such that the resulting tracking performance can compromise with measurement noise, control efforts, and unmodeled system dynamics that exist in practice.

4.3.1. Selection of λ

The design parameter λ determines the control bandwidth of the sliding mode dynamics according to Eq. (25). It is easy to see that a smaller λ leads to a larger bandwidth, indicating a faster response speed and higher tracking accuracy. In Section 3, we have experimentally shown that the dynamic bandwidth of the VCA is considerably higher than that of the DCM. Therefore, we can choose a λ such that the closed-loop bandwidth of the VCA stage is larger than that of the DCM. However, it should be noticed that the benefit of a higher bandwidth comes at the cost of allowing more measurement noise to enter the servo loop which reversely deteriorates the tracking accuracy. For these reasons, we choose $\lambda = 0.0032$ which yields a closed-loop servo bandwidth of 50 Hz for the VCA stage. Comparatively, the closed-loop servo bandwidth of DCM stage is only 10 Hz (see Section 4.2).

4.3.2. Selection of γ

The admissible range for $\gamma \in (1, 2)$ is to avoid the singularity problem [21] in the control inputs. Apparently, a larger γ leads to a smaller convergence time (26) but at the cost of amplifying the measurement noise as well that may consequently introduce extra control noises in u_e (28). Thus, $\gamma = 1.4$ is chosen in our case.

4.3.3. Selection of β

The index $\beta \in (0, 1)$ is actually introduced to smooth the control signals. It can be seen from Eq. (29) that a larger β leads to a smoother control signal u_r and tends to reduce the tracking error according to Eq. (33). Nevertheless, it also results in weaker robustness. Thus, we choose $\beta = 0.8$ for an acceptable balance between robustness and control smoothness.

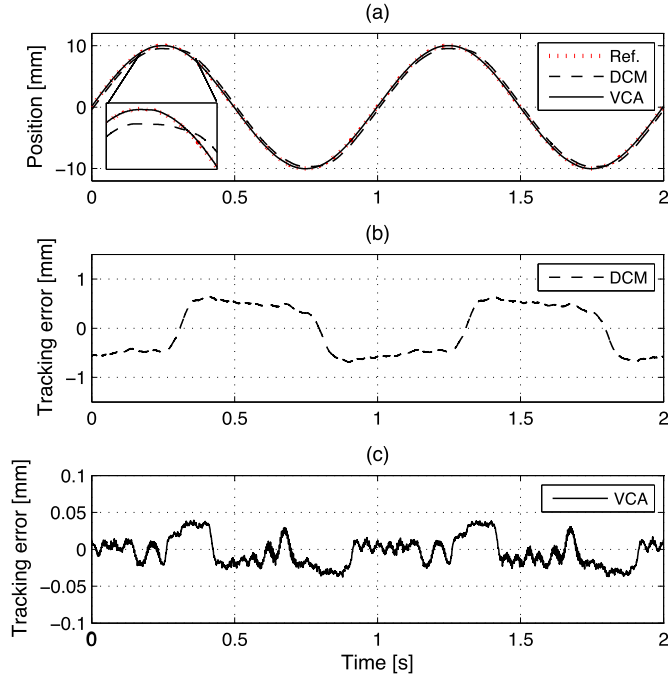


Fig. 4. Tracking performance of the DSA system in the VCA travel range. (a) Position profiles; and tracking errors of (b) DCM; (c) VCA.

4.3.4. Selections of g_1 , g_2

In order to explicitly predict the bound of tracking error from Eqs. (31)–(34), the gains g_1 , g_2 are chosen as time-varying functions given by

$$g_1 = 4000 \left[(\tau - 1) |\ddot{y}_r - \frac{1}{\lambda \gamma} \text{sig}(\dot{e})^{2-\gamma}| + \frac{1}{m_0} \bar{\delta} \right]$$

$$g_2 = 761 \left[(\tau - 1) |\ddot{y}_r - \frac{1}{\lambda \gamma} \text{sig}(\dot{e})^{2-\gamma}| + \frac{1}{m_0} \bar{\delta} \right].$$

Accordingly, the bound of VCA tracking errors can be ideally guaranteed by $|e| \leq 500 \mu\text{m}$.

Note that it can be seen from Eqs. (32) and (33) that a larger gain of either g_1 or g_2 would further reduce the tracking error. However, this comes at the cost of larger control efforts according to Eq. (29), which may cause extra control chattering or even actuator saturation.

4.4. Analysis of avoiding VCA from output saturation

To this end, the control designs for the DCM and VCA stage are addressed. Recall that we also have a design constraint as formulated in Eq. (12), that is, the relative position of the two stages should be within the physical limit of the VCA travel range. Otherwise, the VCA loop will encounter output saturation and may become unstable. Hence, we shall further analyze the condition of Eq. (12) with respect to the given controllers of the two stages and give guidelines on how to retune the controllers to prevent VCA output saturation.

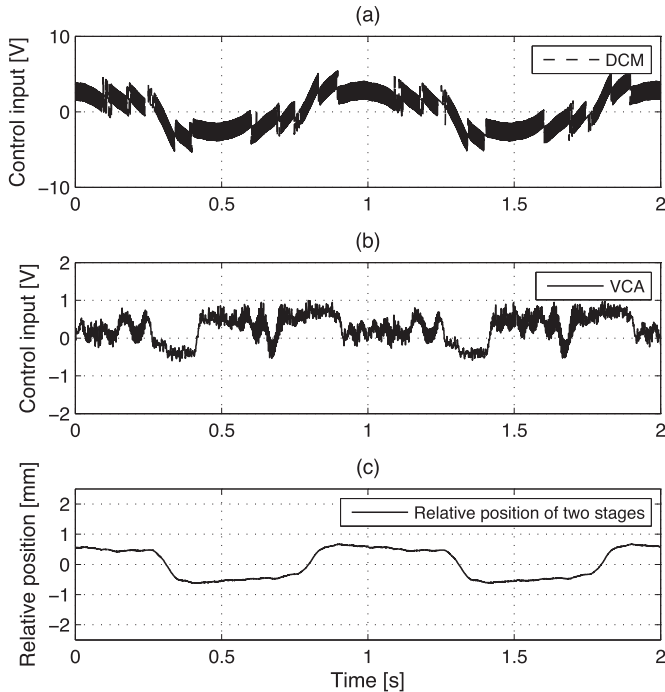


Fig. 5. Relevant signals corresponding to Fig. 4. Control inputs of (a) DCM; (b) VCA; (c) Relative position of the two stages.

Rewrite the relative position of the two stages Δ_y as

$$\begin{aligned}
 \Delta_y &= y - y_1 \\
 &= (y - y_r) - (y_1 - y_r) \\
 &= e - e_1
 \end{aligned} \tag{35}$$

where e_1 and e are the resulting tracking errors under the preceding controllers. Obviously, the condition of Eq. (12) holds provided that

$$|e| + |e_1| \leq L_v \tag{36}$$

because from Eq. (35) we have

$$|\Delta_y| \leq |e| + |e_1| \leq L_v. \tag{37}$$

In our case, the VCA tracking error $|e|$ would be capable of being bounded within the order of micrometers which is negligible as compared to the magnitude of L_v . Therefore, the condition of Eq. (36) is largely determined by the bound of DCM tracking error $|e_1|$. According to Eq. (20), the bound of $|e_1|$ can be reduced by selecting a larger ρ , which is however at the cost of increasing the control input. If this is unacceptable in practice, an alternatively way is to slow down the rate of change of the reference command which could generally decrease the tracking error as well.

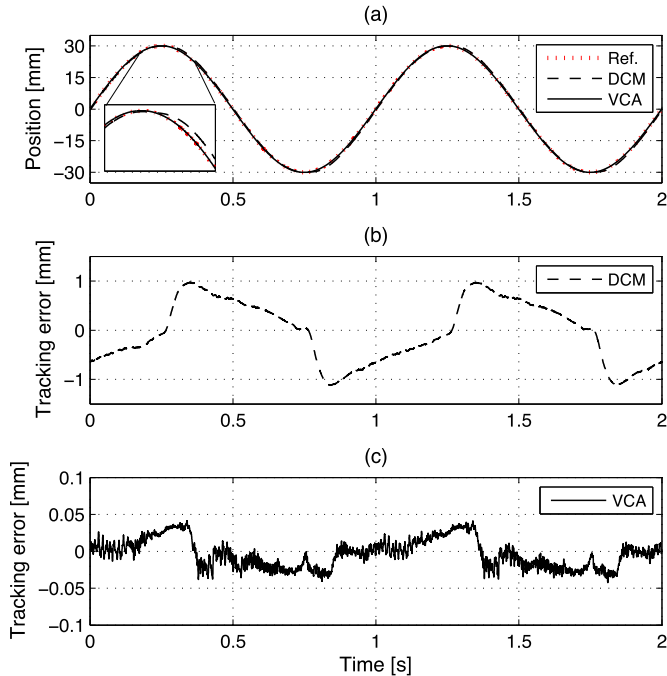


Fig. 6. Tracking performance of the DSA system in the extended range. (a) Position profiles; and tracking errors of (b) DCM; (c) VCA.

5. Experimental results

Experiments are performed on the linear DSA positioning system with the controllers implemented by a real-time DSP system (dSPACE-DS1103) using a sampling period of 0.2 ms.

5.1. Tracking performance in VCA travel range

We first evaluate the tracking performance of the DSA system under the designed controllers in the VCA travel range, i.e., 20 mm. As such, we set the reference command y_r as a sinusoidal waveform with an amplitude of 10 mm. Fig. 4 shows the experimental results from which we can see that the DCM loop can only achieve pretty coarse tracking error around 0.6 mm while the VCA loop further reduces the tracking error (i.e., the overall DSA tracking error) to 40 μm as desired. It should also be noted that both the DCM and VCA loop have endeavored to track the reference simultaneously even when the reference amplitude is in the VCA travel range. In this way, the overall DSA system is able to respond faster and more accurate compared with other sequential control systems with multiple actuators. Furthermore, this feature can keep the relative position of the two stages as small as possible that can be seen in the bottom plot of Fig. 5. The control signals of the DCM and VCA are also shown in Fig. 5(a) and (b), respectively. Although the DCM control signal is quite noisy due to its coarse sensor resolution as well as the differentiation of the sensor signal, it does not deteriorate the DCM control loop significantly. On the other

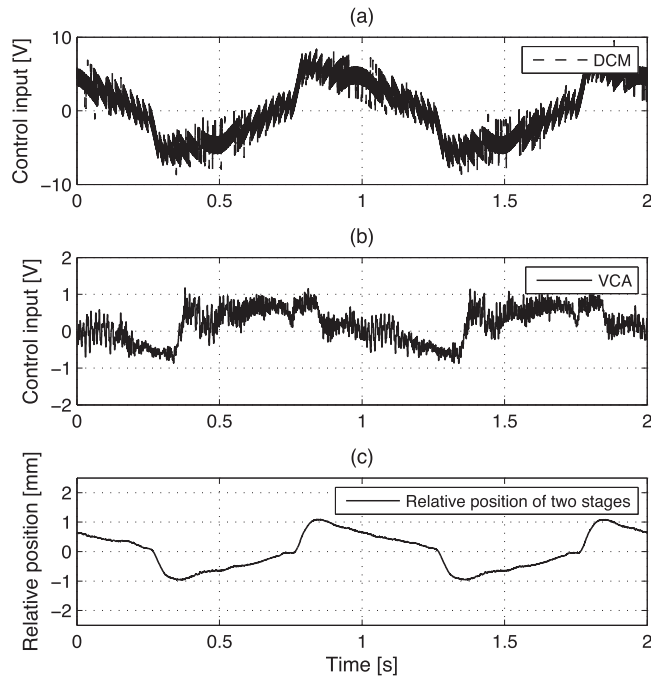


Fig. 7. Relevant signals corresponding to Fig. 6. Control inputs of (a) DCM; (b) VCA; (c) Relative position of the two stages.

hand, the VCA control signal is reasonably smooth owing to the chattering-free FNTSM controller.

5.2. Tracking performance in the extended range

As our main goal is to extend the working range of VCA stage, next we evaluate the tracking performance of the DSA system with the reference amplitude increased to 30 mm. Hence, the VCA is expected to provide a tracking range of 60 mm which is three times its initial travel limit. Fig. 6 shows the tracking performance of the DSA system. We can see from the top plot that the VCA tracks the reference profile with a desired accuracy across the extended range. More specific, the VCA tracking error is still bounded in 40 μm as shown in the bottom plot. Meanwhile, the DCM loop has a much larger tracking error of around 1 mm according to Fig. 6(b). In addition, the error profiles of both the DCM and VCA loop exhibit non-sinusoidal waveforms which result from the residual nonlinear effects such as the friction forces and sensing noises. The corresponding control input signals of the DCM and VCA are also shown in Fig. 7(a) and (b), respectively. We can see that both control input signals contain a certain amount of chattering due to the measurement noises, which however does not cause implementation issues in our setup. Fig. 7(c) shows the relative position of the two stages to verify that the VCA is free of output saturation. The result coincides with the analysis in Section 4.4, which indicates that the relative position is dominated by the tracking error of the DCM loop and is obviously within the VCA travel limit in this case.

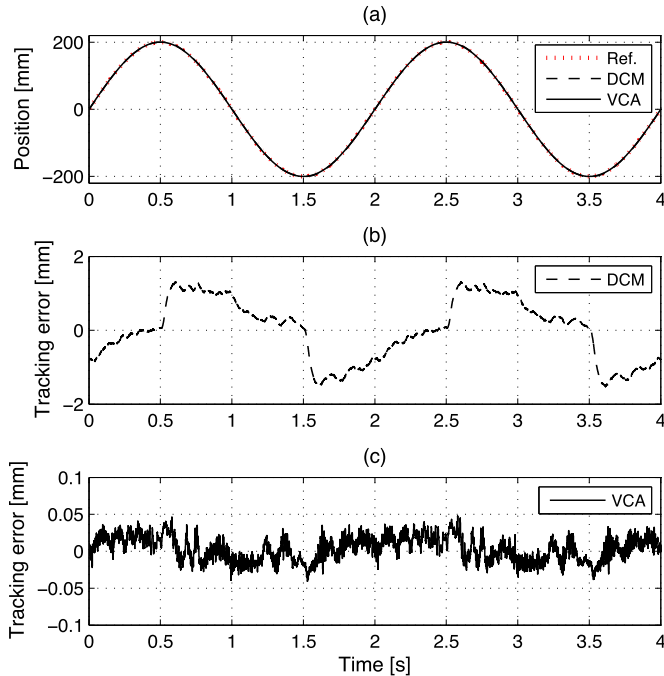


Fig. 8. Tracking performance of the DSA system at the full travel range. (a) Position profiles; and tracking errors of (b) DCM; (c) VCA.

Lastly, the tracking results at the extended full travel range (i.e., 400 mm) are shown in Fig. 8, which also validates the persistent tracking accuracy of 40 μm .

In addition, we carry out the point-to-point tracking control of the DSA system. Such tracking trajectories are commonly used for tasks such as pick-and-place operations. As shown in Fig. 9(a), we employ a third-order minimum jerk trajectory [22] as the reference profile y_r , which is smooth and continuous as well as its corresponding velocity and acceleration trajectories. Fig. 9 shows that the VCA retains the tracking accuracy of 40 μm while the DCM provides the long-range but coarse tracking performance as desired.

5.3. Comparison of FNTSM and H_∞ controller

To demonstrate the benefits of the FNTSM with strong robustness and fast convergence, we also implement an H_∞ controller on the VCA stage for comparison. The H_∞ controller is designed by following the procedure in Section 4.2, where the controller parameters are chosen such that the corresponding VCA closed-loop system (16) has the same bandwidth of 50 Hz as that under the FNTSM controller (see Section 4.3.1) for a fair comparison. In this case, the controller for the DCM stage is unchanged.

Fig. 10 plots the tracking errors of the DSA system in response to the reference profile as shown in Fig. 6(a). Moreover, we consider two cases: without payload (i.e., $\frac{m}{m_0} = 1$); and with payload by placing a 1-kg weight on the VCA stage (i.e., making $\frac{m}{m_0} = 2.7$). It can be seen from Fig. 10 that the tracking error under the H_∞ control becomes much worse when the

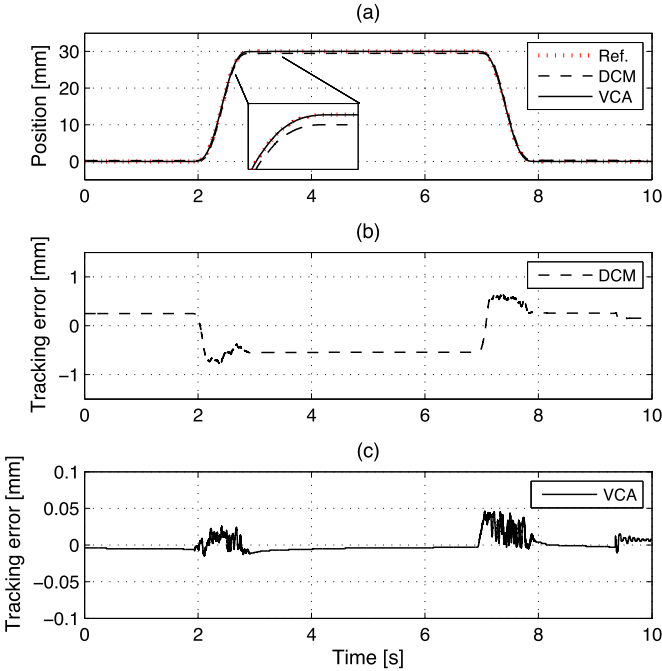


Fig. 9. Point-to-point control performance of the DSA system. (a) Position profiles; and tracking errors of (b) DCM loop; (c) VCA loop.

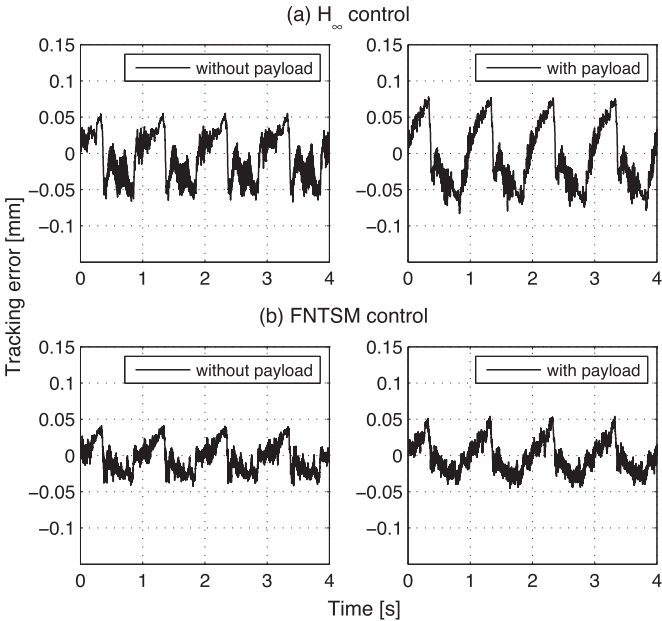


Fig. 10. Comparison of the DSA tracking errors with the VCA stage under (a) H_∞ control; (b) FNTSM control.

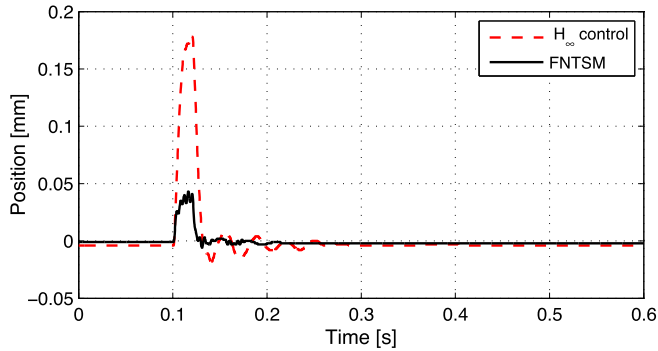


Fig. 11. Performance of external disturbance compensation with the VCA stage under H_∞ and FNTSM control, respectively.

payload is applied. More specific, the tracking error increases from 55 to 80 μm , which is a 45% degradation. Comparatively, the FNTSM control is less affected by the mass uncertainty, where the tracking error changes from 40 to 50 μm indicating a 25% degradation. Therefore, this verifies the stronger robustness of the FNTSM compared with the H_∞ controller.

Lastly, we evaluate the controllers on the compensation for the external shock disturbance acting on the VCA stage. The shock disturbance is with a duration of 20 ms. It is clear from the results in Fig. 11 that the FNTSM not only leads to a smaller amplitude in the position error but also takes significantly less time to settle down around zero. This merit comes from the intrinsic property of finite time convergence in the FNTSM control law.

6. Conclusion

In this paper, we extend the working range of a linear VCA stage from its initial to 400 mm by combining a low-cost DCM stage underneath while guarantee a positioning accuracy of 40 μm . These benefits are achieved by designing an H_∞ controller for the DCM stage and a robust controller based on FNTSM for the VCA stage. The FNTSM controller is capable of achieving fine positioning accuracy and compensating for the payload variation and disturbance. Experimental results demonstrate that the DSA system can not only achieve the desired positioning accuracy across the extended working range but also provide strong robustness against payload variation and faster convergence against external disturbance. Therefore, the developed DSA system enhances the VCA to be applicable to a wider industrial applications that require both high precision and long working range. In the meanwhile, the cost of the overall DSA system is not significantly increased.

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